



SKELETAL & MUSCLE SYSTEM — FULL MIND MAP

Biology Ch. 7.i · Human Anatomy & Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Body Water & Major Chemical Elements
2. Bone Composition & Number of Bones
3. Smallest Bone, Skull, Vertebrae & Ribs
4. Joints, Cartilage & Ligaments
5. Long Bones, Osteoclasts & Arthroplasty
6. Teeth, Dental Plaque & Tooth Enamel
7. Nails, Skin, Hair & Keratin
8. Minerals Required for Bones & Teeth
9. Muscle Working & Important Muscles
10. Important Bone Names — Match the Following

1. Body Water & Major Chemical Elements

- Source ke mutabik human body ka major hissa water se bana hota hai: general range 55–70%; average adult around 55–65%, aur exam questions mein “about 70%” / around 65% bhi diya gaya hai. [Q1, Q2, Q3]
- Human body ka most abundant element Oxygen hai (source around 65% batata hai), mainly body water ki wajah se. [Q4, Q7]
- Approximate body-mass composition source mein Oxygen 65%, Carbon 18.5%, Hydrogen 9.5%, Nitrogen 3.2%, Calcium 1.5% aur Phosphorus 1.0% diya hai. Q5 ke given options mein maximum Calcium hai. [Q5]

⚠ EXAM TRAP — Overall Most Abundant vs “Given Options”

- ▶ Overall human body ka most abundant element Oxygen hai; lekin Q5 ke options (Iron, Sodium, Phosphorus, Calcium) mein Calcium highest hai. [Q4, Q5, Q7]

2. Bone Composition & Number of Bones

- Human bones ka major mineral component calcium phosphate / hydroxyapatite hai; source ke mutabik bone ka about 70% weight modified hydroxyapatite ho sakta hai. [Q6]
- Adult human skeleton mein 206 bones hoti hain. Birth par source about 300 bones batata hai, jo fusion ke baad adulthood tak 206 ho jati hain. [Q8, Q9]
- 20 years ki age par source exact adult count 206 batata hai; Q10 ke numerical options 100, 200, 300 mein se koi correct nahi, isliye “None of these” answer banta hai. [Q10]

3. Smallest Bone, Skull, Vertebrae & Ribs

- Human body ki smallest bone Stapes hai, jo middle ear mein hoti hai. Source saath hi Femur ko largest bone batata hai. [Q11, Q12]
- Source skull count ko 28 bones batata hai: 8 neurocranium + 14 facial bones + 6 middle-ear bones (hyoid excluded). [Q13]
- Human cervical spine mein 7 cervical vertebrae (C1–C7) hoti hain; C1 Atlas aur C2 Axis hain. [Q14]
- Human rib cage mein 12 pairs = total 24 ribs hoti hain; ribs sternum ke saath thoracic organs jaise heart aur lungs ko protection deti hain. [Q17]

4. Joints, Cartilage & Ligaments

- Atlas–Axis (atlantoaxial) joint ko source synovial pivot joint batata hai, jo head rotation allow karta hai. [Q15]

- Q16 mein source neck–head junction ko pivotal / rotary joint ke roop mein frame karta hai, jahan rotational movement hoti hai. [Q16]
- Cartilage bone se softer aur more flexible connective tissue hai; outer ear aur nose-tip examples diye gaye hain. [Q21]
- Ligament dense fibrous connective tissue hai jo two bones ko connect karke joint banane mein help karta hai. [Q22]

5. Long Bones, Osteoclasts & Arthroplasty

- Leg jaise long bone ki diaphysis source ke mutabik hollow-cylinder jaisi hoti hai; inner cavity mein yellow bone marrow hota hai, jabki epiphysis mein sponge-like structure aur red marrow hota hai. [Q18]
- Osteoclast bone tissue ko break down / resorb karne wali bone cell hai aur bone remodeling, maintenance aur pathological bone loss se associated hai. [Q19]

⚠ EXAM TRAP — Q19 — Original Answer-Key Treatment

- ▶ Source explanation ke mutabik Osteoclast ka primary role bone resorption hai, lekin original answer key mein options (a) Bone formation aur (c) Bone fracture/resorption — dono considered bataye gaye hain. Is note ko source ke hisab se preserve kiya gaya hai. [Q19]

- Arthroplasty source mein hip-joint replacement surgery ke roop mein diya gaya hai, jahan damaged hip components ko synthetic implant se replace kiya jata hai. [Q20]

6. Teeth, Dental Plaque & Tooth Enamel

- Dental plaque ek bacterial biofilm / mass hai. Source ke question ke according plaques food particles + saliva + mouth acids + bacteria se bane hote hain; saliva ka average pH 6.7 diya hai. [Q23]
- Human lifetime mein 20 teeth twice develop hote hain. Source mammals ko diphyodont batata hai aur 12 permanent molars ko monophyodont; four tooth types: incisors, canines, premolars, molars. [Q30]
- Tooth enamel source ke mutabik body ka strongest part hai. Ye tooth crown ka hard outer covering hai aur mainly hydroxyapatite (calcium phosphate) crystals se bana hota hai. [Q31]

7. Nails, Skin, Hair & Keratin

- Nails dead tissue / α -keratin ki multiple layers se bane hote hain; nails ke cuttable part mein nerves aur blood vessels nahi hote, isliye cutting par pain feel nahi hota. [Q24]
- Alpha-keratin fibrous structural protein hai jo human skin ki outer epidermal layer ka important material hai; hair aur nails ka bhi major component hai. [Q25]
- Hair keratin protein se bana hota hai; source hair + nails dono mein keratin ko common structural protein batata hai. [Q26, Q27]

8. Minerals Required for Bones & Teeth

- Bones aur teeth ke formation ke liye source Calcium + Phosphorus ko required pair batata hai. [Q28]
- Calcium, phosphorus aur fluorine bones/teeth formation aur strengthening se linked hain; Q29 mein Iodine ko “not essential” answer diya gaya hai, aur iodine ko thyroid function se associate kiya hai. [Q29]

9. Muscle Working & Important Muscles

- Muscles movement ke liye contract/relax karti hain aur source ke mutabik bones ko pull karti hain, push nahi. Muscles pairs mein work karti hain aur contraction par shorter, stiffer aur thicker hoti hain. [Q32]
- Human body ki smallest muscle Stapedius hai, middle ear / tympanic cavity mein. Ye stapes vibration ko stabilize aur regulate karti hai. [Q33]
- Source weight-based strongest muscle Masseter (jaw muscle) batata hai; isliye strongest muscle jaws mein found answer diya hai. [Q34]

10. Important Bone Names — Match the Following

- Correct matching: Breastbone → Sternum | Collarbone → Clavicle | Knee-cap → Patella | Shoulder Blade → Scapula. [Q35]